

Education

Northeastern University	Boston, MA	Expected May 2028
Candidate for Bachelor of Science in Computer Science and Business Administration		GPA: 3.8
Honors/Activities: NU Systematic Alpha, Dean’s List		
Relevant Coursework: Discrete Structures, Introduction to Databases, Program Design & Implementation, Business Statistics, Financial Management		

Technical Skills

Languages: Python, C++, Java, Rust, JavaScript, TypeScript, SQL, Kotlin
Frameworks & Libraries: React, Next.js, Redux, Electron, TensorFlow, Keras, Pandas, NumPy
Developer Tools: Git, AWS (EC2), PostgreSQL, Azure DevOps, IntelliJ, Eclipse, PyCharm, Xcode

Work Experience

Global Risk & Analytics Co-op	December 2025 – June 2026
Wellington Management Boston, MA	
- Built Python risk management tools that apply proprietary factor risk models to compute risk metrics across equity and alternative asset portfolios. - Developed a Model Context Protocol (MCP) server that lets internal LLM workflows query risk metrics, run stress-test scenarios, and flag portfolios breaching mandate limits, adding a compliance guardrail to the risk platform. - Produced top-down portfolio risk reports surfacing concentration risk across multi-asset portfolios.	
Software Engineering Intern	May 2025 – August 2025
Zeal IT Consultants Dallas, TX	
- Developed the frontend for Trinity Industries’ Asset Management System using React and Next.js. - Increased UI development velocity by 10+ story points per sprint and grew team delivery capacity by 300% within one release cycle, accelerating the project timeline by 4 weeks. - Cut page load times by 94% by migrating state management from MobX to Redux and implementing server-side rendering.	

Projects

Event-Driven Implied Volatility Mispricing Python, Polygon.io Report	January 2026 – March 2026
- Built a Python backtesting framework extending Gao, Xing & Zhang (2018) to identify event-driven implied volatility (IV) mispricings across 4 event types on 541 historical S&P 500 constituents and 15 sector ETFs (2016–2025), processing 80,000+ option observations. - Generated sector-level long/short straddle signals achieving a 63% win rate and 3.62% average return per trade out-of-sample on Technology earnings, and a 74% win rate with 10.81% ROI on FOMC events in 2025 testing.	
Prediction Market Trading Rust, AWS Portfolio	December 2025 – Present
- Implemented a mathematical model in Rust that prices a prediction market in real time, hosted on AWS EC2 for low-latency API access. - Identified a market edge and scaled the strategy to a net adjusted Sharpe ratio of 1.2 over two months, with 40% total returns and a maximum drawdown of 10%.	
PM-Trading Desk Python, WebSockets Github	September 2025 – Present
- Developed a prediction market trading application using WebSockets for real-time data access, processing 500+ market updates per second with sub-50ms order execution latency. - Implemented a smart order router spanning Polymarket and Kalshi, achieving 2–5% average price improvement per trade through cross-exchange execution.	
PaveGuard React, Python, YOLO Github	October 2023
- Trained a YOLO image recognition model on 8,000+ images to classify potholes and road fractures across 5 damage types with 71% accuracy. - Built a full-stack crowdsourced reporting platform with a React frontend and Python backend, winning the top prize out of 40+ teams at the AI for All hackathon at UT Dallas.	

Interests

Hackathons, Reading, Rubik’s Cube, Chess, Poker, Baseball, Blogging, Football, Working Out, Watches, Shoes
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